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TITLE PAGE

Problem Statement ID: SIH 26170

Problem Statement Title: AI-Driven Anomaly Detection in Component Burn-In & Screening

Theme: Space Technology

PS Category: Software

Team ID: Nil

Team Name (Registered on portal): Horizon Unbound

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Project Arjuna: Predictive Anomaly Detection

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Proposed Solution: A real-time software tool that continuously ingests multi-channel telemetry (Voltage, Current, Temperature) during component burn-in.

Detailed Explanation

It uses a dual-pipeline AI engine consisting of an Isolation Forest model for sudden multivariate spikes and a CUSUM algorithm for slow thermal drift.

How It Addresses the Problem

It autonomously flags latent degradation and instant faults, halting the screening process before physical thermal runaway damages the components.

Innovation & Uniqueness

It moves beyond traditional static thresholds by capturing micro-fluctuations and thermal slope drift, offering a proactive rather than reactive safety measure.

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TECHNICAL APPROACH

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Technologies Used

- **Languages:** Python 3.x, HTML5, CSS3, Vanilla JavaScript
- **Frameworks & APIs:** FastAPI, WebSockets
- **AI & ML:** scikit-learn (Isolation Forest), CUSUM AI
- **Database:** SQLite
- **Visualization:** Chart.js

Implementation Methodology

Real-time pipeline from hardware simulation to central mission control interface:

01 Hardware Simulation Engine → **02** FastAPI Backend → **03** Isolation Forest & CUSUM AI → **04** SQLite DB & WebSockets → **05** Mission Control UI

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FEASIBILITY AND VIABILITY

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Feasibility Analysis

Highly feasible due to a streamlined, decoupled monolithic architecture that avoids complex microservices, ensuring a 6-day rapid-prototyping completion.

Challenges & Risks

- AI models generating false positives due to normal sensor noise
- Managing persistent WebSocket connections under load

Mitigation Strategies

- Tune Isolation Forest contamination factors and CUSUM thresholds against baseline nominal data to filter out jitter
- Implement event-driven Python architecture with error-handling for connection drops

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IMPACT AND BENEFITS

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Target Audience Impact

Direct Benefit to ISRO

Directly benefits the Indian Space Research Organisation (ISRO) by providing a highly reliable, predictive testing environment for mission-critical space electronics with a sub-100ms detection latency.

Key Benefits

- **Economic:** Prevents the thermal destruction of expensive test sockets and burn-in boards, drastically reducing hardware replacement costs.
- **Operational:** Catches slow degradation 30-60 seconds before static limits are breached, saving components from permanent damage.

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RESEARCH AND REFERENCES

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Space Qualification

Industry Standard

ECSS-Q-ST-60-02C

Space product assurance – ASIC and FPGA development and screening procedures.

Anomaly Detection

Unsupervised ML

Isolation Forest

Liu, F. T., Ting, K. M., & Zhou, Z. H. / Scikit-learn documentation.

Statistical Control

Drift Detection

Continuous Inspection

Page, E. S. 'Continuous Inspection Schemes' (Foundational theory for CUSUM drift detection).